Applied Algebra
Spring 2026
Instructor: Tuan Tran

Exercise Sheet 1

Deadline: by the end of next Tuesday

Instructions. You are encouraged to work in pairs. When submitting your solutions, list the team members and circle the name of the person responsible for writing up the solutions that week. Each week one person should write the solutions, and each student should do so for at least five weeks during the semester.

Please submit written solutions to **TWO** of the following problems.

Problem 1.

(a) Use Euclid's division algorithm to compute $\gcd(4591, 3549)$.

(b) Express $\gcd(4591, 3549)$ as an integer linear combination of 4591 and 3549, i.e., find integers x, y such that $\gcd(4591, 3549) = 4591x + 3549y$.

(c) Explain why your answer in part (b) implies that every common divisor of 4591 and 3549 divides $\gcd(4591, 3549)$.

Problem 2.

(a) Find the prime factorizations of the following integers: 136, 150, 255, 713, 3549.

(b) Use your answers to find the greatest common divisor and least common multiple of each of the pairs: 136 and 150; 255 and 3549.

Problem 3.

(a) Let p be a prime number. Show that if $p \mid a_1 a_2 \cdots a_n$, then $p \mid a_i$ for some i.

(b) Prove that $\sqrt{2}$ is irrational.

(c) Show that if n^2 is divisible by a prime p, then n is divisible by p.

Problem 4.

An L-shaped triomino consists of three unit squares arranged in the shape of the letter "L". Prove by induction that for every integer $n \geq 0$, any $2^n \times 2^n$ checkerboard with one square removed can be tiled completely by L-shaped triominos.